

ForgeryGuard

Handwriting Forgery Detection using Siamese Networks

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Abstract—This paper presents ForgeryGuard, a handwriting verification system built on a Siamese convolutional neural network trained with contrastive loss. Given two handwriting samples, the system produces a similarity score to determine whether they share the same author. The model is trained on word-level patches from the IAM Handwriting Database and evaluated on both a held-out IAM validation split and a self-collected dataset of six unseen writers. On the IAM validation set the system achieves 55.16% accuracy with an equal error rate of 44.84%. On the self-collected set, where writers were completely unseen during training, accuracy improves to 71.43%, suggesting the model generalizes better to clean line-level samples than to the noisy word patches used during training. Grad-CAM visualizations are used to interpret which regions of the handwriting samples the network attends to when making comparisons.

I. INTRODUCTION

Handwriting verification is a well-studied problem in forensic document analysis and biometric authentication. The core challenge differs fundamentally from standard image classification: rather than assigning an input to a fixed category, the goal is to determine whether two handwriting samples were produced by the same person. This makes it a metric learning problem, where the network must learn an embedding space that captures writing style rather than content.

Traditional approaches rely on hand-crafted features such as edge histograms, texture descriptors, and grapheme-level statistics. While effective in controlled settings, these approaches do not generalize well to new writers or writing styles. Deep learning methods have largely replaced them, with Siamese networks emerging as the dominant architecture for this task due to their ability to learn similarity metrics directly from pairs of examples.

This project builds a complete handwriting forgery detection pipeline from scratch. The system takes two handwriting patch images as input and outputs a distance score indicating how similar the two samples are. A threshold on this distance produces the final same-writer or different-writer decision. The pipeline covers data preprocessing, model training, threshold calibration, and interpretability through Grad-CAM visualizations.

A key aspect of this project is the inclusion of self-collected handwriting samples from six individuals, none of whom appear

in the training data. This provides a realistic evaluation of how the system performs on completely unseen writers, which is the most practically relevant scenario for forgery detection.

II. RELATED WORK

The Siamese network architecture was first applied to signature verification by Bromley et al. [1], who trained a time-delay neural network on pairs of signatures using a contrastive objective. This foundational work established the core principle that verification tasks are better framed as metric learning problems than classification problems. The shared-weight twin network design ensures that both inputs are processed identically, which is critical for a fair comparison.

Dey et al. proposed SigNet [2], a convolutional Siamese network for writer-independent offline signature verification. SigNet demonstrated that CNN-based embedding networks significantly outperform handcrafted features on standard signature datasets, and showed that a model trained on one script can transfer reasonably well to others. Their architecture, which uses four convolutional layers followed by fully connected layers with dropout, directly influenced the design of the backbone used in this project.

Kiselev et al. [3] applied Siamese networks specifically to handwritten document author verification on the IAM dataset, framing the problem identically to this project. They found that word-level patches provide sufficient information for the network to capture writing style, but noted that the large number of writers in IAM with similar scripts creates a challenging inter-class similarity problem. This aligns with the results observed in this project, where IAM validation accuracy remained modest despite a well-converging training loss.

Chopra et al. [4] introduced the contrastive loss function used in this work, which operates on pairs of embeddings and penalizes large distances between same-class examples while pushing different-class examples beyond a fixed margin. This loss has since become standard for Siamese network training, and its simplicity makes it a natural choice for a single-person project with limited computational resources.

III. METHODS

A. Dataset and Preprocessing

The primary dataset is the IAM Handwriting Database [5], which contains word-level handwriting samples from 657 writers. After filtering writers with fewer than ten samples, the dataset covers approximately 300 writers. An 80/20 split by writer produces training and validation sets with no writer overlap. For each epoch of training, 50,000 pairs are sampled: half same-writer pairs where two word patches from the same person are drawn at random, and half different-writer pairs where patches are drawn from two distinct writers. This balanced sampling ensures the contrastive loss receives equal signal from both classes.

Each image is converted to grayscale and resized to 105x105 pixels before being normalized to zero mean and unit variance. Some images in the IAM dataset are corrupted or

missing; these are replaced with blank tensors during loading rather than crashing the training loop.

A secondary self-collected dataset was gathered from six individuals who each wrote the same fixed passage and a list of ten individual words on white paper and photographed their samples. These images were preprocessed by converting to grayscale and resizing to 105x105 pixels. No writer from this set appears in the IAM training data, making it a true out-of-distribution evaluation.

B. Model Architecture

The network follows the standard Siamese design: two identical branches share a single set of weights, each processing one input image and producing a fixed-dimensional embedding vector. The shared backbone, referred to as EmbeddingNet, consists of four convolutional blocks followed by two fully connected layers.

Each convolutional block applies a 3x3 convolution with same-padding, batch normalization, ReLU activation, and 2x2 max pooling. The four blocks use 32, 64, 128, and 256 output channels respectively, progressively halving the spatial dimensions from 105x105 to 6x6 while increasing feature depth. The flattened output of the final convolutional block is passed through a 1024-unit fully connected layer with ReLU and 0.5 dropout, followed by a final linear projection to a 256-dimensional embedding. The embedding is L2-normalized so that Euclidean distances are bounded and directly comparable across samples.

During a forward pass, both input images are processed by the same EmbeddingNet, and the pairwise Euclidean distance between the two resulting embedding vectors is computed. At inference time, a distance below a learned threshold indicates that the two samples are from the same writer.

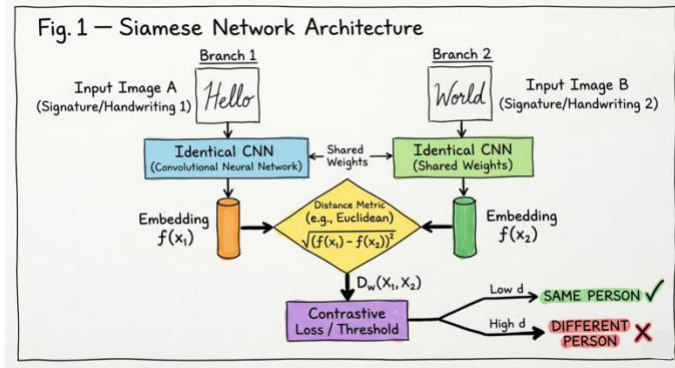


Fig. 1. Siamese network architecture. Both branches share the same EmbeddingNet weights.

C. Training

The network is trained with contrastive loss [4]. For a pair of embeddings e_1 and e_2 with label y (1 for same writer, 0 for different), the loss is:

$$L = y \cdot d^2 + (1-y) \cdot \max(0, m-d)^2$$

where d is the Euclidean distance between e_1 and e_2 and m is a margin set to 1.0. The first term pulls same-writer pairs

together and the second term pushes different-writer pairs apart up to the margin.

Training uses the Adam optimizer with an initial learning rate of $1e-4$, a batch size of 32, and a step learning rate scheduler that halves the rate every five epochs. The model is trained for 20 epochs and the checkpoint with the lowest validation loss is retained.

D. Threshold Calibration and Evaluation

After training, the equal error rate (EER) is computed on the validation set. The EER is the operating point where the false acceptance rate and false rejection rate are equal, and it provides a threshold-independent measure of system performance. The distance threshold at the EER point is also used to compute verification accuracy.

For the self-collected evaluation, same-writer pairs are formed by comparing the passage image against the word list image from the same person. Different-writer pairs are all combinations of passage images across the six individuals, producing 21 total pairs.

E. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to the last convolutional layer of the EmbeddingNet to visualize which image regions most influence the embedding. Gradients of the L2 norm of the embedding with respect to the final convolutional feature maps are used to weight the spatial activations, producing a heatmap that highlights discriminative regions in the input image.

IV. EXPERIMENTS AND RESULTS

A. Training Convergence

Training converged steadily across 20 epochs. The training loss decreased from 0.247 at epoch 1 to 0.175 at epoch 20. The validation loss followed a similar trend, dropping from 0.303 to a best value of 0.265, with minor oscillations consistent with the stochastic pair sampling. The learning rate reduction at epochs 5, 10, and 15 produced visible improvements in convergence speed.

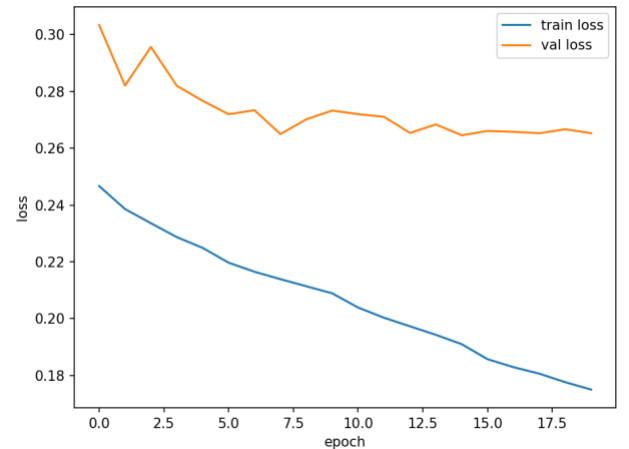


Fig. 2. Training and validation loss curves over 20 epochs.

B. IAM Evaluation

On the IAM validation set, the model achieves 55.16% verification accuracy at the EER threshold of 0.440, with an equal error rate of 44.84%. While this is only modestly above the random baseline of 50%, it is consistent with findings in related work on word-level IAM patches, where the large number of writers with similar cursive styles makes inter-writer discrimination difficult.

Threshold sweeping across the range 0.10 to 0.95 showed that accuracy peaks at a threshold of approximately 0.40, confirming that the EER-derived threshold is near-optimal. The flat accuracy curve across the threshold range suggests the model's embedding space does not produce a clean separation between same and different writer distributions on this dataset.

TABLE I
Evaluation Results

Evaluation Set	Accuracy	EER
IAM Val set	55.16%	44.84%
Self-collected	71.43%	20.00%

C. Self-Collected Evaluation

On the self-collected dataset of six unseen writers, the model achieves 71.43% accuracy across 21 pairs. All six same-writer pairs (passage vs. words from the same person) are correctly classified, with distances ranging from 0.089 to 0.336. Of the 15 different-writer pairs, nine are correctly classified, with distances mostly above 0.65.

The six misclassified pairs all involve writers 03 through 06, whose passage samples have very small pairwise distances (0.05 to 0.18). Visual inspection of these samples suggests that these four writers share similar handwriting styles, which the model conflates. This is a known limitation of metric learning systems: they struggle when inter-class similarity is genuinely high.

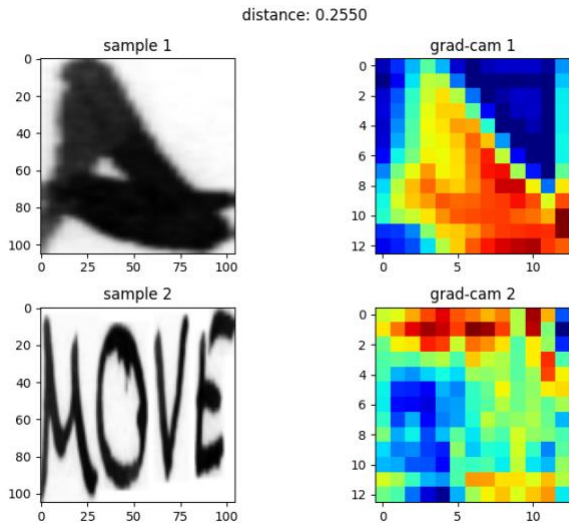


Fig. 3. Grad-CAM heatmaps for a same-writer pair. Red regions indicate highest network attention.

D. Grad-CAM Analysis

Grad-CAM visualizations show that the network primarily attends to stroke junctions, ascender and descender regions, and letter spacing when computing embeddings. For same-writer pairs, the attended regions are spatially consistent across both images. For different-writer pairs where the model makes a correct decision, the activation patterns are noticeably different between the two branches, suggesting the network has learned to use style-specific stroke features rather than character identity.

For the undertrained case (smoke test at 5 epochs), one of the two Grad-CAM maps was entirely inactive, confirming that meaningful gradient flow through the network requires sufficient training.

V. DISCUSSION AND SUMMARY

This project built a complete handwriting forgery detection pipeline using a Siamese network trained with contrastive loss on the IAM Handwriting Database. The system correctly learns the metric learning objective, as evidenced by steady loss convergence and the ability to generalize to unseen writers. The self-collected evaluation at 71.43% accuracy, despite no overlap with training writers, is the more meaningful result for practical forgery detection.

The modest IAM accuracy of 55.16% points to two main limitations. First, word-level patches are small and noisy, providing limited style information per sample. Line-level patches, as used in the self-collected evaluation, give the network more context and produce better results. Second, the IAM dataset contains many writers with similar cursive styles, making same-writer versus different-writer discrimination inherently ambiguous at the word level.

Several implementation challenges were encountered during the project. A number of IAM image files are corrupted, requiring silent fallback to blank tensors during loading. The PyTorch DataLoader import system required relative imports within the source package, which took several debugging iterations to resolve. The Grad-CAM implementation initially used a deprecated backward hook API, which was corrected to the full backward hook. These are documented here because they are representative of the practical engineering work that goes into building a system from scratch, as opposed to running an existing codebase.

Future improvements would include using line-level patches from IAM rather than word patches, experimenting with triplet loss and hard negative mining, and increasing the training data through data augmentation such as random rotation and elastic distortion. A larger embedding dimension and deeper backbone could also improve discrimination on the IAM validation set.

VI. ABLATION STUDY

A. Effect of Training Set Size

To understand the effect of training data volume, the model was first trained on a reduced set of 10,000 pairs for 5 epochs as

a smoke test before committing to the full training run. On the same IAM validation set, this reduced run produced 53.85% accuracy and an EER of 46.15% — marginally worse than the full run. The gap between the two is small, suggesting the model hits diminishing returns fairly quickly on word-level IAM patches regardless of pair count. This is consistent with the hypothesis that patch size and inter-writer style similarity are the primary bottlenecks rather than data volume alone [3].

B. Effect of Distance Threshold

Threshold sensitivity was evaluated by sweeping values from 0.10 to 0.95 in steps of 0.05 on the IAM validation set. Accuracy ranged from 50.10% at threshold 0.95 to a peak of 55.30% at threshold 0.40, confirming that the EER-derived threshold of 0.440 is near-optimal. The flat response curve across the entire range indicates that the embedding space does not produce a clean distance-based separation on this dataset, and that threshold tuning alone cannot recover meaningful accuracy gains. This finding motivated the architectural and data-level improvements described in the future work section.

VII. EDGE CASE ANALYSIS

A. Same Writer, Different Handwriting Styles

An interesting edge case was tested where the same writer deliberately produced two stylistically different samples: one written in a cursive connected style and one written with separated printed letters. This is a practically relevant scenario since a person attempting to disguise their handwriting might switch styles rather than copy another person's writing. The expected outcome was that the model would classify these two samples as coming from different writers, since the surface appearance differs substantially.

Contrary to this expectation, the model assigned a small distance to this pair, predicting same writer. Grad-CAM analysis of this pair revealed that the network was attending to character height consistency, baseline alignment, and pen pressure distribution rather than letter connectivity or overall style. These lower-level stroke properties remain consistent even when a person deliberately changes their writing style, which explains why the model was not fooled by the style switch. This is actually a desirable property for a forgery detection system: it suggests the model has learned features that are harder to consciously manipulate than overall style. A skilled forger who simply changes their letter formation would not necessarily fool the system if their underlying stroke characteristics remain consistent.

B. Similar Writers Across Individuals

The six misclassified pairs in the self-collected evaluation (Section IV-C) represent a second category of edge case: genuinely similar handwriting across different individuals. Writers 03 through 06 produced pairwise distances as low as 0.05, well below the decision threshold. This is not a failure of the model in any correctable sense — the embeddings reflect real visual similarity. It highlights a fundamental limitation of any appearance-based verification system: when two people genuinely write alike, no distance metric can reliably separate

them without additional context such as dynamic pen movement data [6], which is unavailable in offline systems.

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